

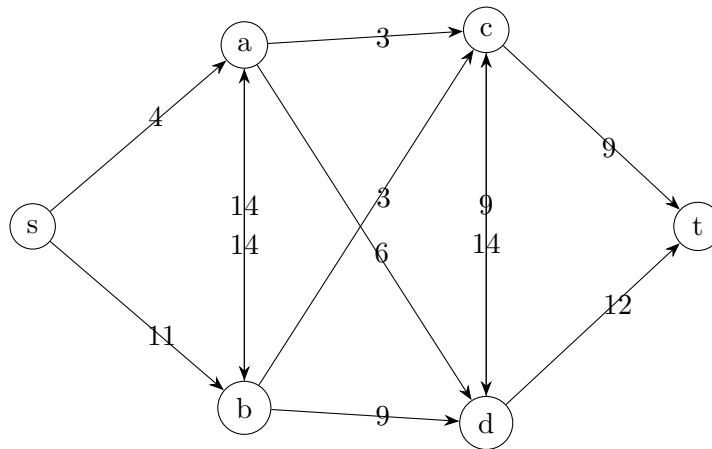
CS 253: Data Structures and Algorithms

Randomized Final Study Problem

Graphs

Ford-Fulkerson Maximum Flow

Starting with an initial flow of 0 on every edge, perform the Ford-Fulkerson algorithm to find the maximum flow from s to t . For each iteration, show the augmenting path, its bottleneck capacity, and the updated flow values.



Instructor Reference

Maximum flow value: 15. Augmenting path sequence: $s \rightarrow a \rightarrow c \rightarrow t$ (bottleneck 3); $s \rightarrow a \rightarrow d \rightarrow t$ (bottleneck 1); $s \rightarrow b \rightarrow c \rightarrow t$ (bottleneck 3); $s \rightarrow b \rightarrow d \rightarrow t$ (bottleneck 8). Final edge flows (flow/capacity): $s \rightarrow a$: 4/4, $s \rightarrow b$: 11/11, $a \rightarrow b$: 0/14, $b \rightarrow a$: 0/14, $a \rightarrow c$: 3/3, $a \rightarrow d$: 1/6, $b \rightarrow c$: 3/3, $b \rightarrow d$: 8/9, $c \rightarrow d$: 0/14, $d \rightarrow c$: 0/9, $c \rightarrow t$: 6/9, $d \rightarrow t$: 9/12.